

Package ‘CrossingTools’

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Title Tools for Cross Optimization in Plant Breeding Programs

Version 1.1.0

Description Mate allocation is a critical component of plant breeding programs, aiming to generate superior progenies and improved populations over time. While numerous methodologies for mate allocation exist, few software tools offer a unified and scalable framework that integrates these methodologies in a user-friendly manner. 'CrossingTools' addresses this gap by providing a flexible, efficient solution for optimizing mating decisions across a wide range of breeding schemes. The package supports genomic BLUP and selection indices for various variance structures and implements multiple selection criteria to evaluate cross potential, including expected mean, optimal haploid value (OHV), and superior progeny value (SPV). Different methods are provided for calculating additive and dominance segregation variances for both inbred and heterozygous parents, making 'CrossingTools' suitable for a wide range of breeding strategies. After criterion calculation, users can rank potential crosses or apply the built-in optimal crossing selection routine, which balances genetic gain and diversity to support long-term improvement. The package combines R's scripting flexibility with high-performance C++ routines via 'Rcpp' and 'RcppArmadillo', ensuring scalability for large populations with dense marker data.

License GPL-3

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LinkingTo Rcpp,
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backsolve_marker_effects

Backsolve marker effects from individual genetic effects

Description

Backsolves marker effects from predicted individual genetic effects using the equivalence between genomic relationship-matrix and marker-effect models (Stranden and Garrick, 2009).

Usage

```
backsolve_marker_effects(marker.mat, G.mat, genotype.effects)
```

Arguments

marker.mat	Numeric marker design matrix with individuals in rows and markers in columns. Its coding, centring, and marker representation must correspond to those used to construct G.mat.
G.mat	Numeric square genomic relationship matrix among the individuals in marker.mat.
genotype.effects	Numeric vector, matrix, or data frame containing predicted individual genetic effects. Rows must correspond to individuals in marker.mat; columns represent traits.

Details

Let M denote the marker design matrix, G the genomic relationship matrix, and g_t the vector of predicted individual genetic effects for trait t . Marker effects are first obtained as

$$\tilde{\mu}_t = M^\top G^{-1} g_t.$$

Because genomic relationship matrices may differ from $MM^\top$ by a constant scaling factor, the backsolved marker effects are subsequently rescaled so that the genetic values reconstructed from the marker effects optimally reproduce the supplied individual genetic effects. For each trait, the scalar

$$c_t = \frac{(M\tilde{\mu}_t)^\top g_t}{(M\tilde{\mu}_t)^\top (M\tilde{\mu}_t)}$$

is calculated, and the final marker effects are

$$\mu_t = c_t \tilde{\mu}_t.$$

Thus, scaling is estimated independently for each trait and corresponds to a single constant applied to all marker effects within that trait. This is appropriate when `G.mat` is proportional to the crossproduct of the supplied marker matrix, for example

$$G = \frac{MM^\top}{s},$$

where s is a scalar normalization constant.

Genomic relationship matrices using marker-specific or other non-scalar scaling are not supported.

The marker matrix supplied to the function must use the same coding, centring, and marker representation as that used to construct `G.mat`. For example, centred additive marker genotypes can be used to recover average allele-substitution effects from genomic breeding values when `G.mat` was constructed from the same centred additive marker matrix (VanRaden, 2008; Strandén and Garrick, 2009). Likewise, dominance effects can be backsolved by supplying the corresponding dominance design matrix and dominance genomic relationship matrix.

Multiple traits can be processed simultaneously by supplying `genotype.effects` as a matrix or data frame with one column per trait. Marker effects and their scalar rescaling factors are calculated separately for each trait using the same marker and genomic relationship matrices.

Value

A data frame with markers in rows and traits in columns containing the backsolved marker effects.

References

- VanRaden, P. M. (2008). Efficient methods to compute genomic predictions. *Journal of Dairy Science*, 91(11), 4414–4423. doi:10.3168/jds.20070980
- Strandén, I. and Garrick, D. J. (2009). Derivation of equivalent computing algorithms for genomic predictions and reliabilities of animal merit. *Journal of Dairy Science*, 92(6), 2971–2975. doi:10.3168/jds.20081929

calc_midparent_inbred *Calculate expected breeding values for inbred crosses*

Description

Calculates the expected genomic breeding value of progeny from proposed biparental or four-parent crosses among inbred lines.

Usage

```
calc_midparent_inbred(
  crosses,
  marker.mat,
  marker.effects,
  weights = NULL,
  p = NULL,
  nthreads = 4L
)
```

Arguments

crosses	Matrix or data frame with two columns for biparental crosses or four columns for four-parent crosses. Parent identifiers may be row indices of marker.mat or character identifiers matching rownames(marker.mat).
marker.mat	Numeric marker dosage matrix with individuals in rows and markers in columns, coded 0, 1, and 2 for the counted allele.
marker.effects	Numeric matrix of average allele-substitution effects (α) with markers in rows and traits in columns. Effects should be parameterised relative to the reference population defined by p. If p = NULL, the reference allele frequencies are derived from marker.mat. Its number of rows must equal ncol(marker.mat).
weights	Optional numeric vector with one weight per trait. When supplied, a weighted genomic breeding-value index is calculated for each cross.
p	Optional numeric vector of reference allele frequencies, with one value per marker. These frequencies define the reference population used to centre marker genotypes for breeding-value calculations. If NULL, allele frequencies are calculated from marker.mat.
nthreads	Positive integer. Number of computational threads.

Details

Under an additive genetic model, the expected breeding value of progeny from a biparental cross equals the average of the parental breeding values, i.e. the mid-parent value (Falconer and Mackay, 2009). For four-parent crosses, the expected breeding value is analogously obtained as the average breeding value of the four contributing parents.

Breeding values are calculated from centred marker genotypes, $M - 2p$, and average allele-substitution effects (α), where p contains the allele frequencies of the reference population. The reference allele frequencies can be supplied by the user. If they are not supplied, they are calculated from marker.mat. Thus, p defines the reference population relative to which breeding values are expressed.

Multiple traits can be evaluated simultaneously. Optional trait weights can be used to calculate a weighted breeding-value index for each cross.

Value

If `weights = NULL`, a data frame containing the parental identifiers followed by one GEBV.<trait> column per trait.

If `weights` is supplied, a list containing:

`cross.df` The parental identifiers and trait-specific genomic breeding values.

`index.df` The parental identifiers and the weighted index GEBV.IDX.

References

Falconer, D. S. and Mackay, T. (2009). *Introduction to Quantitative Genetics*. 4th ed. Pearson, Prentice Hall, Harlow.

calc_midparent_outcross

Calculate expected breeding and total genetic values for biparental outcrosses

Description

Calculates the expected genomic breeding value and expected total genetic value of proposed two-way crosses from additive and dominance marker effects.

Usage

```
calc_midparent_outcross(
  crosses,
  marker.mat,
  marker.effects.A,
  marker.effects.D = NULL,
  weights = NULL,
  p = NULL,
  nthreads = 4L
)
```

Arguments

<code>crosses</code>	Matrix or data frame with two columns specifying the parents of each proposed cross. Parent identifiers may be row indices of <code>marker.mat</code> or character identifiers matching <code>rownames(marker.mat)</code> .
<code>marker.mat</code>	Numeric marker dosage matrix with individuals in rows and markers in columns, coded 0, 1, and 2 for the counted allele.
<code>marker.effects.A</code>	Numeric matrix of average allele-substitution effects (α) with markers in rows and traits in columns. Effects should be parameterised relative to the reference population defined by <code>p</code> . If <code>p = NULL</code> , the reference allele frequencies are derived from <code>marker.mat</code> . Its number of rows must equal <code>ncol(marker.mat)</code> .

marker.effects.D	Optional numeric matrix of dominance marker effects (d) with markers in rows and traits in columns, used with the statistical dominance-deviation parameterisation. It must have the same dimensions as marker.effects.A. If NULL, dominance effects are assumed to be zero.
weights	Optional numeric vector with one weight per trait. When supplied, weighted indices are calculated from the trait-specific GEBVs and TGVs.
p	Optional numeric vector of reference allele frequencies, with one value per marker. These frequencies define the reference population used to centre marker genotypes and parameterise statistical dominance deviations. If NULL, allele frequencies are calculated from marker.mat.
nthreads	Positive integer. Number of computational threads.

Details

Under an additive genetic model, the expected breeding value of progeny from a cross between two individuals equals the average of the parental breeding values, i.e. the mid-parent value (Falconer and Mackay, 2009). Breeding values are calculated from centred marker genotypes, $M - 2p$, and average allele-substitution effects (α), where p contains the allele frequencies of the reference population.

When dominance effects are included, the expected total genetic value (TGV) additionally accounts for the expected statistical dominance deviation of the F1 progeny. This allows crosses between complementary parents to be distinguished when their expected performance differs because of dominance, which can be relevant for the selection of crossing parents in outcrossing and clonally propagated breeding programmes (Werner et al., 2023).

The allele frequencies in p define the reference population used for the statistical additive and dominance parameterisation. If p is not supplied, allele frequencies are calculated from marker.mat. The supplied average allele-substitution and dominance effects should therefore correspond to the same reference population.

Multiple traits can be evaluated simultaneously. Optional trait weights can be used to calculate indices for the expected breeding value and expected total genetic value.

Value

If weights = NULL, a data frame containing the parental identifiers followed by GEBV.<trait> and TGV.<trait> for each trait.

If weights is supplied, a list containing:

cross.df The parental identifiers and trait-specific GEBVs and TGVs.

index.df The parental identifiers and the weighted indices GEBV.IDX and TGV.IDX.

References

- Falconer, D. S. and Mackay, T. (2009). *Introduction to Quantitative Genetics*. 4th ed. Pearson, Prentice Hall, Harlow.
- Werner, C. R., Gaynor, R. C., Sargent, D. J., Lillo, A., Gorjanc, G. and Hickey, J. M. (2023). Genomic selection strategies for clonally propagated crops. *Theoretical and Applied Genetics*, 136, 74. doi:10.1007/s00122023043006

calc_optimal_haploid_value

Calculate optimal haploid values for candidate crosses

Description

Calculates the Optimal Haploid Value (OHV) of proposed biparental or four-parent crosses from additive marker effects.

Usage

```
calc_optimal_haploid_value(
  crosses,
  marker.mat,
  marker.effects,
  weights = NULL,
  haplotype.blocks = NULL,
  nthreads = 4L
)
```

Arguments

crosses	Matrix or data frame with two columns for biparental crosses or four columns for four-parent crosses. Parent identifiers may be integer row indices of <code>marker.mat</code> or character identifiers matching <code>rownames(marker.mat)</code> .
marker.mat	Numeric marker matrix with individuals in rows and markers in columns.
marker.effects	Numeric matrix of additive marker effects with markers in rows and traits in columns. Its number of rows must equal <code>ncol(marker.mat)</code> .
weights	Optional numeric vector with one weight per trait. When supplied, a weighted OHV index is calculated as the linear combination of the trait-specific OHVs.
haplotype.blocks	Optional data frame defining the haplotype blocks. It must contain: <code>block</code> Block identifier. Markers with the same identifier belong to the same haplotype block. <code>site</code> Marker identifier given as a column index of <code>marker.mat</code> or as a marker name matching <code>colnames(marker.mat)</code>. Each marker may occur in at most one block. If <code>NULL</code> , each marker is treated as an independent block.
nthreads	Positive integer. Number of computational threads.

Details

The OHV describes the maximum additive genetic value obtainable by combining favourable parental haplotypes and was proposed as a criterion for selecting breeding material according to its potential to generate superior fixed progeny (Daetwyler et al., 2015). Related haplotype-based approaches have also been used to characterise Mendelian sampling effects and upper selection limits (Cole and VanRaden, 2011).

For each haplotype block, the function determines the most favourable parental contribution according to the supplied marker effects and sums these contributions across blocks. If `haplotype.blocks`

= NULL, each marker is treated as an independent block. User-defined haplotype blocks can instead be supplied to account for sets of markers that should be inherited jointly.

The OHV represents an upper bound on additive cross potential rather than the expected mean performance of progeny. In contrast to criteria such as the superior progeny value (SPV), it does not use the expected progeny mean and segregation variance to describe the distribution of progeny values. Instead, it evaluates the most favourable combination of parental genetic material that can be assembled under the specified haplotype-block definition.

Multiple traits can be evaluated simultaneously. If trait weights are supplied, trait-specific OHVs are combined into a weighted OHV index.

A practical consideration is that certain plant-breeding procedures based on the OHV criterion have been the subject of patent protection, including US11744199B2 (Daetwyler et al., 2023). Patent scope and status may differ among jurisdictions and over time; users intending commercial application should assess whether relevant patent claims apply to their intended use.

Value

If weights = NULL, a data frame containing the original cross definitions followed by one OHV.<trait> column per trait.

If weights is supplied, a list containing:

- cross.df The original cross definitions and the trait-specific OHVs.
- index.df The cross definitions and the weighted OHV index in column OHV.IDX.

References

Cole, J. B. and VanRaden, P. M. (2011). Use of haplotypes to estimate Mendelian sampling effects and selection limits. *Journal of Animal Breeding and Genetics*, 128(6), 446–455. doi:10.1111/j.14390388.2011.00922.x

Daetwyler, H. D., Hayden, M. J., Spangenberg, G. C. and Hayes, B. J. (2015). Selection on Optimal Haploid Value increases genetic gain and preserves more genetic diversity relative to genomic selection. *Genetics*, 200(4), 1341–1348. doi:10.1534/genetics.115.178038

Daetwyler, H. D., Hayes, B. J., Robbins, K., Hayden, M. J. and Spangenberg, G. (2023). Selection based on optimal haploid value to create elite lines. U.S. Patent US11744199B2.

calc_spv_inbred	Calculate segregation variance and superior progeny values for inbred lines
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Description

Calculates expected genomic breeding values, segregation variances, and superior progeny values for doubled haploid or recombinant inbred line families derived from proposed two-way or four-way crosses.

Usage

```
calc_spv_inbred(
  crosses,
  genetic.map,
  marker.mat,
  marker.effects,
  t,
  intensity = 1,
  type = "DH",
  weights = NULL,
  p = NULL,
  covariance = FALSE,
  method = 1,
  nthreads = 4L
)
```

Arguments

<code>crosses</code>	Matrix or data frame with two columns for two-way crosses or four columns for four-way crosses. Parent identifiers may be row indices of <code>marker.mat</code> or character identifiers matching <code>rownames(marker.mat)</code> .
<code>genetic.map</code>	Data frame describing the genetic map. It must contain: <code>site</code> Marker index or marker name matching <code>colnames(marker.mat)</code> . <code>chr</code> Chromosome identifier. <code>pos</code> Marker position on the chromosome in Morgan. All markers in <code>marker.mat</code> must be represented.
<code>marker.mat</code>	Numeric marker dosage matrix with individuals in rows and markers in columns, coded 0, 1, and 2 for the counted allele.
<code>marker.effects</code>	Numeric matrix of average allele-substitution effects (α) with markers in rows and traits in columns. Effects should be parameterised relative to the reference population defined by <code>p</code> . If <code>p = NULL</code> , the reference allele frequencies are derived from <code>marker.mat</code> . Its number of rows must equal <code>ncol(marker.mat)</code> .
<code>t</code>	Integer. Number of additional generations considered before final line development. Its interpretation depends on the selected offspring type and segregation-variance method.
<code>intensity</code>	Numeric scalar giving the standardised selection intensity used to calculate superior progeny values. The default is 1.
<code>type</code>	Character. Offspring type, either "DH" for doubled haploid lines or "RIL" for recombinant inbred lines.
<code>weights</code>	Optional numeric vector with one weight per trait. When supplied, weighted index values are calculated for each cross.
<code>p</code>	Optional numeric vector of reference allele frequencies, with one value per marker. These frequencies define the reference population used to centre marker genotypes for breeding-value calculations. If <code>NULL</code> , allele frequencies are calculated from <code>marker.mat</code> .
<code>covariance</code>	Logical. If <code>TRUE</code> , also calculate segregation covariance matrices among traits for each cross.

method	Method used to calculate segregation variance. Either 1 or "lehermeier" for the Lehermeier et al. (2017) method, or 2 or "osthushenrich" for the Osthushenrich et al. (2017) method. The latter is currently available only for two-way crosses. Four-way crosses are calculated using the multi-parental formulation of Allier et al. (2019).
nthreads	Positive integer. Number of computational threads.

Details

The calculation can account for additional generations of random mating or inbreeding before line development. Multiple traits and optional weighted trait indices are supported.

Segregation (co)variances are calculated analytically from parental marker genotypes, recombination fractions, and average allele-substitution effects. For biparental crosses, the available methods follow the analytical approaches of Lehermeier et al. (2017) and Osthushenrich et al. (2017). Both provide expressions for doubled haploid (DH) and recombinant inbred line (RIL) populations after an arbitrary number of intermating generations. Four-way crosses are calculated using the extension to multi-parental crosses described by Allier et al. (2019).

Additive genetic values are expressed as breeding values using centred marker genotypes, $M - 2p$, and average allele-substitution effects (α). The reference allele frequencies p can be supplied by the user. If they are not supplied, they are calculated from marker.mat. Thus, p defines the reference population relative to which breeding values are expressed.

Superior progeny values (SPV) combine the expected family breeding value with the predicted segregation standard deviation and a user-supplied selection intensity. For multiple traits, segregation covariance matrices can additionally be calculated and combined with trait weights to obtain an index-level SPV.

Value

If neither weights nor trait covariances are requested, a data frame containing the parental identifiers and, for each trait, GEBV.<trait>, var.<trait>, and SPV.<trait>.

If weights is supplied, a list additionally containing index.df, with columns GEBV.IDX, var.IDX, and SPV.IDX.

If covariance = TRUE, a list containing:

cross.df Cross-specific trait values and segregation variances.

index.df Weighted index values, if weights is supplied.

covariances Segregation covariance matrices for each cross.

References

- Lehermeier, C., Teyssedre, S. and Schon, C.-C. (2017). Genetic gain increases by applying the usefulness criterion with improved variance prediction in selection of crosses. *Genetics*, 207(4), 1651–1661. doi:10.1534/genetics.117.300403
- Osthushenrich, T., Frisch, M. and Herzog, E. (2017). Genomic selection of crossing partners on basis of the expected mean and variance of their derived lines. *PLOS ONE*, 12(12), e0188839. doi:10.1371/journal.pone.0188839
- Allier, A., Moreau, L., Charcosset, A., Teyssedre, S. and Lehermeier, C. (2019). Usefulness criterion and post-selection parental contributions in multi-parental crosses: Application to polygenic trait introgression. *G3: Genes|Genomes|Genetics*, 9(5), 1469–1479. doi:10.1534/g3.119.400129

calc_spv_outcross	<i>Calculate segregation variance and superior progeny values for outcrosses</i>
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Description

Calculates expected genomic breeding values, expected total genetic values, additive and dominance segregation variances, and superior progeny values for proposed F1 crosses.

Usage

```
calc_spv_outcross(
  crosses,
  genetic.map,
  hap.mat1,
  hap.mat2,
  marker.effects.A,
  marker.effects.D,
  intensity = 1,
  weights = NULL,
  p = NULL,
  covariance = FALSE,
  nthreads = 4L
)
```

Arguments

crosses	Matrix or data frame with two columns specifying the parents of each proposed cross. Parent identifiers may be row indices of hap.mat1 and hap.mat2, or character identifiers matching their row names.
genetic.map	Data frame describing the genetic map. It must contain: site Marker identifier given as a column index or as a marker name matching colnames(hap.mat1). chr Chromosome identifier. pos Marker position on the chromosome in Morgan. All markers in the haplotype matrices must be represented.
hap.mat1	Numeric haplotype matrix with individuals in rows and markers in columns, containing the first haplotype of each individual.
hap.mat2	Numeric haplotype matrix with the same dimensions as hap.mat1, containing the second haplotype of each individual.
marker.effects.A	Numeric matrix of average allele-substitution effects (α) with markers in rows and traits in columns. Effects should be parameterised relative to the reference population defined by p. If p = NULL, the reference allele frequencies are derived from hap.mat1 and hap.mat2. Its number of rows must equal ncol(hap.mat1).
marker.effects.D	Numeric matrix of dominance marker effects (δ) with markers in rows and traits in columns, used with the statistical dominance-deviation parameterisation. It must have the same dimensions as marker.effects.A.

intensity	Numeric scalar giving the standardized selection intensity used to calculate SPV and TSPV. The default is 1.
weights	Optional numeric vector with one weight per trait. When supplied, weighted index values are calculated for each cross.
p	Optional numeric vector of reference allele frequencies, with one value per marker. If NULL, allele frequencies are calculated from hap.mat1 and hap.mat2. The supplied frequencies are used to centre marker genotypes for breeding-value calculations.
covariance	Logical. If TRUE, also calculate breeding-value, dominance-deviation, and combined additive–dominance segregation covariance matrices among traits for each cross.
nthreads	Positive integer. Number of computational threads.

Details

Additive superior progeny values are calculated from the breeding-value segregation variance, whereas total superior progeny values are calculated from the total genetic segregation variance, including breeding-value variance, dominance-deviation variance, and their covariance.

Segregation (co)variances are calculated analytically from phased parental haplotypes, recombination fractions, and additive and dominance marker effects, following the framework of Bonk et al. (2016) for Mendelian sampling covariability among full-sib progeny.

Additive genetic values are expressed as breeding values using centred marker genotypes, $M - 2p$, and average allele-substitution effects (α). Dominance is represented by statistical dominance deviations. The expected total genetic value is therefore the sum of the expected breeding value and expected dominance deviation.

For multiple traits, the corresponding segregation covariance matrices are calculated analogously. These quantities can be combined with a selection intensity to obtain additive superior progeny values (SPV) and total superior progeny values (TSPV).

Value

If neither weights nor trait covariances are requested, a data frame containing the parental identifiers and, for each trait, GEBV.<trait>, TGV.<trait>, var.A.<trait>, SPV.<trait>, var.D.<trait>, var.TGV.<trait>, and TSPV.<trait>.

If weights is supplied, index.df contains GEBV.IDX, TGV.IDX, var.A.IDX, SPV.IDX, var.D.IDX, var.TGV.IDX, and TSPV.IDX. The total genetic segregation variance var.TGV.IDX includes breeding-value variance, dominance-deviation variance, and their covariance.

If covariance = TRUE, a list containing:

cross.df Cross-specific trait values and segregation variances.

index.df Weighted index values, if weights is supplied.

additive.covariances Breeding-value segregation covariance matrices for each cross.

dominance.covariances Dominance-deviation segregation covariance matrices for each cross.

additive.dominance.covariances Combined additive–dominance covariance matrices, $\Sigma_{AD} + \Sigma_{DA}$, for each cross.

References

Bonk, S., Reichelt, M., Teuscher, F., Segelke, D. and Reinsch, N. (2016). Mendelian sampling covariability of marker effects and genetic values. *Genetics Selection Evolution*, 48, 36. doi:10.1186/s1271101602140

make_cross_plan	<i>Create a crossing plan from candidate parents</i>
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Description

Generates all possible crosses among a supplied set of candidate parents.

Usage

```
make_cross_plan(
  parents = NULL,
  self = FALSE,
  male.parents = NULL,
  female.parents = NULL
)
```

Arguments

parents	Optional vector of parent identifiers for an unsexed crossing design. At least two unique identifiers must be supplied.
self	Logical. If TRUE, include self-crosses in an unsexed crossing design. The argument is ignored when male.parents and female.parents are used.
male.parents	Optional vector of male parent identifiers for a sexed crossing design.
female.parents	Optional vector of female parent identifiers for a sexed crossing design.

Details

For an unsexed crossing design, all unique unordered pairwise combinations of the supplied parents are returned. Self-crosses can optionally be added. Reciprocal crosses are not distinguished; for example, the crosses A x B and B x A are represented by a single combination.

For a sexed crossing design, all combinations between the supplied male and female parents are returned. Reciprocal crosses are therefore distinguished when the same parents occur in both sets. The self argument is not applied to sexed designs; consequently, a self-cross is included whenever the same parent occurs in both male.parents and female.parents.

Either parents or both male.parents and female.parents should be supplied. If parents is supplied together with sex-specific parent sets, the latter are ignored.

Value

A data frame with one row per cross. For unsexed designs, the columns are parent1 and parent2. For sexed designs, the columns are male and female.

make_index

*Calculate a multi-trait selection index***Description**

Constructs a multi-trait selection index from genomic predicted effects using either a Desired Gains Index or user-supplied coefficients for a Smith–Hazel index.

Usage

```
make_index(
  genotype.effects,
  weights = NULL,
  var.mat = NULL,
  desired.gains = NULL,
  nthreads = 4L
)
```

Arguments

genotype.effects	Numeric matrix or data frame containing predicted genotype effects, with genotypes in rows and traits in columns.
weights	Optional numeric vector with one index coefficient per trait. When supplied, a Smith–Hazel-type linear selection index is calculated as the weighted sum of the trait effects. The supplied values are interpreted directly as index coefficients.
var.mat	Optional numeric covariance matrix used to construct the Desired Gains Index. It may be either an $n_{\text{trait}} \times n_{\text{trait}}$ trait covariance matrix or the covariance matrix of the stacked multi-trait genotype effects. If NULL, the trait covariance matrix is estimated from genotype.effects.
desired.gains	Optional numeric vector specifying the desired relative response for each trait. The relative values define the desired direction of multi-trait genetic improvement.
nthreads	Positive integer. Number of computational threads.

Details

Linear selection indices combine information from multiple traits into a single criterion on which individuals can be ranked and selected (Smith, 1936; Hazel, 1943). When weights are supplied, the index for individual i is calculated as

$$I_i = b^\top g_i,$$

where g_i is the vector of predicted effects for individual i and b contains the supplied index coefficients. This corresponds to the linear-index formulation underlying the Smith–Hazel selection index. The coefficients are supplied directly and are not estimated from economic weights and separate phenotypic and genetic covariance matrices.

Alternatively, a Desired Gains Index can be constructed by specifying the desired relative response for each trait. The Desired Gains Index provides a way to express the breeding objective directly in

terms of the relative changes desired among traits, rather than specifying index coefficients directly (Werner et al., 2024).

When `desired.gains` is supplied, the index coefficients are derived from the desired-gain vector and the covariance structure of the predicted genotype effects. The covariance information can be supplied either as a trait covariance matrix or as the covariance matrix of the stacked multi-trait genotype effects. If `var.mat = NULL`, the trait covariance matrix is estimated from `genotype.effects`. Exactly one of `desired.gains` or `weights` must be supplied.

Value

A list containing:

`index` A data frame containing the selection-index value for each genotype.

`weights` The index coefficients used to construct the selection index. For a Desired Gains Index, these are derived from the desired gains and covariance structure; otherwise they are the supplied coefficients.

References

- Smith, H. F. (1936). A discriminant function for plant selection. *Annals of Eugenics*, 7(3), 240–250.
- Hazel, L. N. (1943). The genetic basis for constructing selection indexes. *Genetics*, 28(6), 476–490.
- Hazel, L. N. and Lush, J. L. (1942). The efficiency of three methods of selection. *Journal of Heredity*, 33(11), 393–399. [doi:10.1093/oxfordjournals.jhered.a105102](https://doi.org/10.1093/oxfordjournals.jhered.a105102)
- Pesek, J. and Baker, R. J. (1969). Comparison of tandem and index selection in the modified pedigree method of breeding self-pollinated species. *Canadian Journal of Plant Science*, 49(6), 773–781. [doi:10.4141/cjps69132](https://doi.org/10.4141/cjps69132)
- Werner, C. R., Gardner, K. A. and Tolhurst, D. J. (2024). Reviving the Desired Gains Index: An optimal solution for parent selection in public plant breeding programs. *bioRxiv*, 2024.07.21.603926. [doi:10.1101/2024.07.21.603926](https://doi.org/10.1101/2024.07.21.603926)

optimize_cross_plan	<i>Optimise crossing plans for genetic gain and diversity</i>
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Description

Selects a crossing plan of `ncrosses` crosses by balancing a user-defined cross criterion against genomic similarity among the contributing parents. Fixed crosses and upper and lower constraints on parental contributions can be incorporated directly into the optimisation.

Usage

```
optimize_cross_plan(
  candidate.crosses,
  fixed.crosses = NULL,
  ncrosses,
  target.angle = 0,
  criterion,
  criterion.fixed = NULL,
  G.mat,
```

```

    parents.upper = NA,
    parents.lower = NA,
    method = "pareto",
    return.params = FALSE,
    plot = TRUE,
    params = list(),
    nthreads = 4L
)

```

Arguments

<code>candidate.crosses</code>	Integer or character matrix or data frame with two columns. Candidate crosses from which the variable part of the crossing plan is selected. Self-crosses are not allowed.
<code>fixed.crosses</code>	Integer or character matrix or data frame with two columns, or NULL. Crosses included in every returned plan. These must use the same type of parent identifiers as <code>candidate.crosses</code> . Self-crosses are not allowed.
<code>ncrosses</code>	Integer. Total number of crosses in each returned plan, including fixed crosses. Must be at least <code>nrow(fixed.crosses)</code> .
<code>target.angle</code>	Numeric scalar between 0 and 90. Target trade-off angle used when <code>method = "angle"</code> . An angle of 0 gives maximum emphasis to the cross criterion, whereas an angle of 90 gives maximum emphasis to reducing genomic similarity.
<code>criterion</code>	Numeric vector with one cross-criterion value for each row of <code>candidate.crosses</code> . Larger values are considered favourable.
<code>criterion.fixed</code>	Numeric vector with one cross-criterion value for each row of <code>fixed.crosses</code> , or NULL when no fixed crosses are used.
<code>G.mat</code>	Numeric square matrix containing genomic relationships, similarities, or another pairwise measure used to quantify similarity among parental contributions. Character parent identifiers require matching row names.
<code>parents.upper</code>	Numeric vector with one value per parent in <code>G.mat</code> , giving the maximum number of times each parent may occur across the complete crossing plan. Each cross contributes one occurrence to each parent. Values must be non-negative whole numbers or Inf. The default is Inf for every parent.
<code>parents.lower</code>	Numeric vector with one value per parent in <code>G.mat</code> , giving the minimum number of times each parent must occur across the complete crossing plan. Each cross contributes one occurrence to each parent. Values must be non-negative whole numbers and cannot exceed the corresponding values in <code>parents.upper</code> . The default is zero for every parent.
<code>method</code>	Character string specifying the optimisation method. Must be either <code>"angle"</code> or <code>"pareto"</code> .
<code>return.params</code>	Logical. Whether to return optimisation diagnostics in addition to the selected crossing plan or Pareto solutions.
<code>plot</code>	Logical. Whether to display the estimated Pareto frontier when <code>method = "pareto"</code> . Ignored when <code>method = "angle"</code> .
<code>params</code>	Named list of genetic-algorithm parameters. Unspecified entries use their default values: <code>probability.mutate</code> Numeric scalar in $[0, 1]$. Mutation probability for each offspring.

<code>n.mutate</code>	Non-negative integer. Number of positions changed when mutation occurs.
<code>n.select</code>	Integer of at least 2. Number of solutions selected in each generation.
<code>n.pop</code>	Integer of at least 2. Population size.
<code>max.generation</code>	Positive integer. Maximum number of generations.
<code>max.iteration</code>	Positive integer. Number of generations without improvement after which optimisation stops.
<code>angle.penalty</code>	Non-negative numeric scalar controlling the penalty for deviation from <code>target.angle</code> , used only when <code>method = "angle"</code> .
<code>nthreads</code>	Positive integer. Number of computational threads.

Details

Ranking crosses solely according to their predicted value may maximise short-term genetic gain but can increase coancestry among selected parents and reduce genetic diversity. Optimal cross selection therefore combines expected cross performance with a measure of relatedness among the parents contributing to a crossing plan.

For each candidate crossing plan, performance is quantified by the mean value of the supplied cross criterion across all selected crosses. Genomic similarity is calculated from proportional parental contributions and the relationship matrix supplied in `G.mat`, analogous to group coancestry as used in optimal contribution selection (Meuwissen, 1997). Each mating contributes equally to the crossing plan, with half of its contribution assigned to each parent.

Because the cross criterion and genomic similarity may be measured on different scales, the objectives are normalised before optimisation. CrossingTools provides two strategies for balancing the resulting gain–diversity objectives. With `method = "angle"`, the user specifies a target angle between 0 and 90 degrees. An angle of 0 degrees gives maximum emphasis to the cross criterion, whereas an angle of 90 degrees gives maximum emphasis to reducing genomic similarity and therefore maintaining diversity. Intermediate angles specify different compromises between the two objectives.

With `method = "pareto"`, the cross criterion and genomic similarity are treated as separate objectives and optimised simultaneously using an NSGA-II-type multi-objective genetic algorithm (Deb et al., 2002). The returned non-dominated solutions approximate the Pareto frontier and represent alternative trade-offs between predicted cross performance and genomic diversity.

Both optimisation strategies search over complete crossing plans rather than ranking crosses independently. Practical breeding constraints can be imposed through fixed crosses and minimum or maximum parental contributions. Crosses that should not be considered can be excluded from `candidate.crosses`.

Value

For `method = "angle"` and `return.params = FALSE`, a data frame with columns `parent1` and `parent2` describing the selected crossing plan.

For `method = "angle"` and `return.params = TRUE`, a list containing:

`crossPlan` The selected crossing plan as a data frame with columns `parent1` and `parent2`.

`uMax`, `simMax` The maximum attainable cross criterion and the corresponding similarity.

`uBest`, `simBest` The cross criterion and similarity of the selected solution.

`angleBest`, `lenBest` The angle and length of the selected solution in normalised objective space.

For `method = "pareto"`, a list containing:

`pareto.plans` A list of Pareto-optimal crossing plans, each represented by a data frame with columns `parent1` and `parent2`.

`pareto.frontier` A data frame containing the solution identifier `pareto.id`, cross criterion `u`, and genomic similarity `sim`.

References

Meuwissen, T. H. E. (1997). Maximizing the response of selection with a predefined rate of in-breeding. *Journal of Animal Science*, 75(4), 934–940.

Deb, K., Pratap, A., Agarwal, S. and Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2), 182–197. [doi:10.1109/4235.996017](https://doi.org/10.1109/4235.996017)

<code>plot_cross_plan</code>	<i>Plot predicted progeny distributions for a crossing plan</i>
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Description

Visualises the predicted progeny distributions and available cross criteria for the crosses in a crossing plan. Separate panels are produced for each trait.

Usage

```
plot_cross_plan(cross.plan, cross.df, traits = NULL, nsamples = 1000L)
```

Arguments

<code>cross.plan</code>	Data frame, matrix, or object coercible to a data frame containing the crossing plan. It must have two columns for biparental crosses or four columns for four-parent crosses. Rows determine the displayed cross labels and order.
<code>cross.df</code>	Data frame, matrix, or object coercible to a data frame containing cross-specific predictions, typically produced by functions for calculating segregation variance, cross expectation, SPV, or OHV. Trait-specific columns must follow the naming convention <code><metric>.<trait></code> , for example <code>GEBV.Yield</code> , <code>SPV.Yield</code> , <code>var.A.Yield</code> , or <code>var.TGV.Yield</code> . Biparental crosses are matched using columns <code>parent1</code> and <code>parent2</code> , male and female, or the first two columns. Four-parent crosses require columns <code>parent1</code> to <code>parent4</code> .
<code>traits</code>	Character vector specifying the trait names to plot, or <code>NULL</code> to plot all traits detected in <code>cross.df</code> .
<code>nsamples</code>	Positive integer. Number of random samples drawn per cross, trait, and distribution type to visualise the predicted distributions. The default is 1000.

Details

Predicted progeny distributions are represented by normal distributions defined by the expected cross value and the corresponding segregation variance. The displayed densities are generated by random sampling from normal distributions with the predicted means and variances for visualisation only; they do not represent simulated progeny or meioses.

For additive genetic values, the progeny distribution is defined by the expected genomic breeding value (GEBV) and additive segregation variance. When dominance is included, the total genetic distribution is defined by the expected total genetic value (TGV) and total genetic segregation variance, including additive variance, dominance variance, and the additive–dominance covariance contribution.

Available cross criteria are plotted on the distributions as points. These can include the expected genomic breeding value (GEBV), expected total genetic value (TGV), superior progeny value (SPV), total superior progeny value (TSPV), and optimal haploid value (OHV). Only criteria present in `cross.df` are displayed.

Value

A ggplot object containing trait-specific predicted progeny distributions and available cross-criterion values.

predict_response	<i>Predict response to multi-trait index selection</i>
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Description

Calculates the expected response to multi-trait selection for either a desired-gains index or an index with user-defined trait weights.

Usage

```
predict_response(
  var.mat = NULL,
  desired.gains = NULL,
  weights = NULL,
  intensity = 1,
  plot = TRUE
)
```

Arguments

var.mat	Numeric square covariance matrix. It may either be an $n_{\text{trait}} \times n_{\text{trait}}$ trait covariance matrix or a block-structured covariance matrix containing one $n_{\text{trait}} \times n_{\text{trait}}$ diagonal block per genotype. In the latter case, the diagonal blocks are averaged to obtain the trait covariance matrix used for index calculations.
desired.gains	Optional numeric vector specifying the desired relative response for each trait. Index weights are calculated from the desired response vector and the trait covariance matrix as $b = \Sigma^{-1}g$.
weights	Optional numeric vector specifying the fixed Smith–Hazel index coefficients for each trait.
intensity	Numeric scalar giving the standardised selection intensity used to scale the expected responses. The default is 1.
plot	Logical. If TRUE, plot the expected response for each trait. The default is TRUE.

Details

Linear selection indices combine information from multiple traits into a single selection criterion and provide a framework for predicting correlated responses to selection (Smith, 1936; Hazel, 1943). Index selection can improve multiple traits simultaneously and has long been applied in plant and animal breeding (Hazel and Lush, 1942; Pesek and Baker, 1969).

For a desired-gains index, index weights are derived from a vector of desired responses g as

$$b = \Sigma^{-1}g,$$

where Σ is the trait covariance matrix. The desired-gains formulation specifies the relative direction of genetic improvement rather than fixing the absolute magnitude of response. Consequently, the predicted responses are proportional to the supplied desired gains, while their overall magnitude depends on the covariance structure and the selection intensity. This formulation allows breeding objectives to be expressed directly as desired relative changes among traits without requiring explicit economic weights (Werner et al., 2024).

Alternatively, fixed index weights can be supplied directly. This corresponds to the classical linear selection-index framework of Smith (1936) and Hazel (1943), in which multiple traits are combined into a single selection criterion. Here, the supplied weights are treated directly as the index coefficients b . For a vector of index coefficients b , the index variance is

$$\sigma_I^2 = b^\top \Sigma b,$$

and the expected response in trait k is

$$\Delta G_k = i \frac{(\Sigma b)_k}{\sqrt{b^\top \Sigma b}},$$

where i is the selection intensity. The function additionally returns the correlation between each trait and the selection index.

Exactly one of `desired.gains` or `weights` must be supplied.

If `var.mat` is a block-structured covariance matrix containing one trait covariance block per genotype, the diagonal trait covariance blocks are averaged before constructing the index. Covariances between different genotypes are therefore not used in the response calculation.

Value

A list containing:

`overall.df` A data frame containing the index variance `index.var` and expected response in the index `gain.index`.

`trait.df` A data frame containing, for each trait, its variance, correlation with the index, expected response, and index weight.

References

- Smith, H. F. (1936). A discriminant function for plant selection. *Annals of Eugenics*, 7(3), 240–250.
- Hazel, L. N. (1943). The genetic basis for constructing selection indexes. *Genetics*, 28(6), 476–490.
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Werner, C. R., Gardner, K. A. and Tolhurst, D. J. (2024). Reviving the Desired Gains Index: An optimal solution for parent selection in public plant breeding programs. *bioRxiv*, 2024.07.21.603926. doi:10.1101/2024.07.21.603926

summarize_cross_plan	<i>Summarize the expected performance of a crossing plan</i>
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Description

Calculates the mean predicted performance of the crosses in `cross.plan` for each available trait and value metric, including GEBV, TGV, SPV, TSPV, and OHV. Crosses are matched to `cross.df` using their parental identifiers.

Usage

```
summarize_cross_plan(  
  cross.plan,  
  cross.df,  
  plot = T,  
  reference.pop = NA,  
  weights = NA  
)
```

Arguments

<code>cross.plan</code>	Data frame, matrix, or object coercible to a data frame containing the crossing plan. It must have two columns for two-way crosses or four columns for four-way crosses.
<code>cross.df</code>	Data frame, matrix, or object coercible to a data frame containing predictions for the candidate crosses. Trait-specific value columns must follow the naming convention <code><metric>.<trait></code> , for example <code>GEBV.Yield</code> or <code>SPV.Height</code> . Crosses are matched using columns <code>parent1</code> and <code>parent2</code> , male and female, or the first two columns. Four-way crosses require columns <code>parent1</code> to <code>parent4</code> .
<code>plot</code>	Logical. If TRUE, plot the predicted mean GEBV response.
<code>reference.pop</code>	Optional numeric matrix or data frame with individuals in rows and traits in columns. When supplied, predicted responses are expressed relative to the corresponding trait means. It must contain at least two individuals and all traits represented in <code>cross.df</code> .
<code>weights</code>	Optional numeric vector of trait weights, in the same order as the traits in the returned summary. When supplied together with <code>reference.pop</code> , it is used to calculate and display the desired response direction.

Details

Optionally, the mean GEBV response can be plotted. When `reference.pop` is supplied, responses are expressed relative to the reference-population means. When trait weights are also supplied, the corresponding desired-response direction is shown for comparison.

Value

A data frame with one row per trait. The output contains:

`trait` Trait identifier parsed from the prediction-column names.

`ncrosses` Number of crosses in `cross.plan`.

`mean.GEBV`, `mean.TGV`, `mean.SPV`, `mean.TSPV`, `mean.OHV` Mean value of each available metric across the selected crosses. Metrics not available for a particular trait are returned as NA.

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